

Classification and Ulam stability of the absolute radical equation

Candidate full answer to Question 1 of arXiv:2007.01454

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Status. Candidate full answer, likely valid, pending independent expert and novelty review. It classifies all algebraic solutions for every positive integer n and proves uniform Hyers–Ulam stability in every real Banach target, with an explicit constant.

1 Question and notation

Fix $n \in \mathbb{N}$ and nonzero $a, b, c, d \in \mathbb{R}$. The source asks for the solution and Ulam stability of

$$f\left(\sqrt[n]{|ax^n + by^n|}\right) + f\left(\sqrt[n]{|ax^n - by^n|}\right) = cf(x) + df(y), \quad x, y \in \mathbb{R}. \quad (1)$$

Question 1. Considered as a future work concerning the solution and Ulam's stability for the following functional equation:

$$f\left(\sqrt[n]{|ax^n + by^n|}\right) + f\left(\sqrt[n]{|ax^n - by^n|}\right) = cf(x) + df(y), \quad x, y \in \mathbb{R},$$

where a, b, c and d are nonzero real constants and n is a positive integer.

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Put $A = |a|$ and $B = |b|$. A mapping $Q : \mathbb{R} \rightarrow V$ into a real vector space is called quadratic when

$$Q(u + v) + Q(u - v) = 2Q(u) + 2Q(v) \quad (u, v \in \mathbb{R}). \quad (2)$$

Equivalently, $Q(t) = S(t, t)$ for a symmetric biadditive map $S : \mathbb{R}^2 \rightarrow V$. In particular Q is even.

2 Complete algebraic classification

Theorem 1 (All exact solutions). *Let V be any real vector space. A function $f : \mathbb{R} \rightarrow V$ satisfies (1) if and only if*

$$f(x) = Q(x^n) + w \quad (x \in \mathbb{R}), \quad (3)$$

where $Q : \mathbb{R} \rightarrow V$ is quadratic, $w \in V$, and

$$2Q(At) = cQ(t), \quad 2Q(Bt) = dQ(t) \quad (t \in \mathbb{R}), \quad (4)$$

together with

$$(2 - c - d)w = 0. \quad (5)$$

Proof. Changing x to $-x$ leaves the unordered pair of the two terms on the left of (1) unchanged: for even n the terms are individually unchanged, and for odd n they are interchanged. Since $c \neq 0$, every solution is even. The same conclusion also follows by changing y .

For $s \geq 0$ set

$$g(s) = f(s^{1/n}).$$

For $s, t \geq 0$, the unordered pair $\{|as + bt|, |as - bt|\}$ equals $\{As + Bt, |As - Bt|\}$. Thus (1) becomes

$$g(As + Bt) + g(|As - Bt|) = cg(s) + dg(t). \quad (6)$$

Let $w = g(0)$ and $h = g - w$. Putting $s = t = 0$ gives (5). Putting in turn $t = 0$ and $s = 0$, and using (5), gives

$$2h(As) = ch(s), \quad 2h(Bt) = dh(t). \quad (7)$$

Now substitute $s = u/A$, $t = v/B$ in (6) and eliminate the two right-hand terms using (7). One obtains

$$h(u + v) + h(|u - v|) = 2h(u) + 2h(v) \quad (u, v \geq 0). \quad (8)$$

The even extension $Q(r) = h(|r|)$ satisfies (2), because the unordered pair $\{|x + y|, |x - y|\}$ equals $\{|x| + |y|, ||x| - |y||\}$. Equation (7) extends by evenness to (4). Finally, evenness of f gives $f(x) = Q(x^n) + w$ for every sign of x .

Conversely, let (3)–(5) hold. Evenness of Q removes both absolute values, (2) changes the left side into $2Q(ax^n) + 2Q(by^n) + 2w$, and (4)–(5) turn this into the right side of (1). $\square$

Corollary 2 (Continuous solutions). *If V is a real topological vector space and f is continuous, then*

$$f(x) = x^{2n}v + w$$

for $v, w \in V$ satisfying

$$(2A^2 - c)v = (2B^2 - d)v = (2 - c - d)w = 0.$$

In a real Banach space, the same conclusion follows from any standard regularity assumption forcing a quadratic Cauchy map to be continuous.

Proof. The quadratic map in Theorem 1 is continuous and hence $Q(t) = t^2Q(1)$. Substitute this into (4). $\square$

3 Uniform Hyers–Ulam stability

For a map $f : \mathbb{R} \rightarrow Y$, define its defect $D_f(x, y)$ as the left side of (1) minus $cf(x) + df(y)$.

Theorem 3 (Explicit Ulam bound). *Let Y be a real Banach space and suppose*

$$\|D_f(x, y)\| \leq \varepsilon \quad (x, y \in \mathbb{R}). \quad (9)$$

Put $M = \max\{|c|, |d|\}$. There is an exact solution $F(x) = Q(x^n) + w$ of (1) such that

$$\sup_{x \in \mathbb{R}} \|f(x) - F(x)\| \leq \frac{4}{3}\varepsilon + \frac{2}{M}\varepsilon + \begin{cases} 0, & c + d = 2, \\ \frac{\varepsilon}{|2 - c - d|}, & c + d \neq 2. \end{cases} \quad (10)$$

One can take $w = f(0)$ in the resonant case $c + d = 2$, and $w = 0$ otherwise. The quadratic part is explicitly

$$Q(t) = \lim_{k \rightarrow \infty} 4^{-k} \left[f\left((2^k|t|)^{1/n}\right) - f(0) \right]. \quad (11)$$

Proof. As in the exact proof, changing one sign leaves the radical sum unchanged. Comparing the two defects gives

$$\|f(x) - f(-x)\| \leq \frac{2\varepsilon}{M}. \quad (12)$$

Define $g(s) = f(s^{1/n})$ for $s \geq 0$, put $g_0 = g(0)$ and $h = g - g_0$. The transformed defect is

$$E(s, t) = g(As + Bt) + g(|As - Bt|) - cg(s) - dg(t), \quad \|E(s, t)\| \leq \varepsilon. \quad (13)$$

For $u, v \geq 0$ set

$$R_A(u) = E(u/A, 0), \quad R_B(v) = E(0, v/B), \quad R_0 = E(0, 0).$$

Direct cancellation gives the key identity

$$\begin{aligned} h(u+v) + h(|u-v|) - 2h(u) - 2h(v) \\ = E(u/A, v/B) - R_A(u) - R_B(v) + R_0. \end{aligned} \quad (14)$$

Hence the norm of the left side is at most 4ε .

Extend h evenly to $H : \mathbb{R} \rightarrow Y$ by $H(t) = h(|t|)$. Then $H(0) = 0$ and

$$\|H(x+y) + H(x-y) - 2H(x) - 2H(y)\| \leq 4\varepsilon.$$

Putting $y = x$ shows that $4^{-k}H(2^k t)$ is Cauchy. Its limit is the Q in (11); the telescoping series gives

$$\|H(t) - Q(t)\| \leq \sum_{k=0}^{\infty} \frac{4\varepsilon}{4^{k+1}} = \frac{4}{3}\varepsilon. \quad (15)$$

Dividing the approximate quadratic identity at $(2^k x, 2^k y)$ by 4^k and passing to the limit proves that Q is exactly quadratic.

The one-variable defects in (13) also give

$$\|2h(As) - ch(s)\| \leq 2\varepsilon, \quad \|2h(Bs) - dh(s)\| \leq 2\varepsilon.$$

Apply these at $2^k s$, divide by 4^k , and pass to the limit. Thus Q satisfies the exact scaling laws (4).

If $c + d = 2$, choose $w = g_0$. Otherwise choose $w = 0$; since $\|(2 - c - d)g_0\| \leq \varepsilon$, we have $\|g_0 - w\| \leq \varepsilon/|2 - c - d|$. Theorem 1 shows that $F(x) = Q(x^n) + w$ is exact. For $x \geq 0$, the asserted distance follows from (15) and the bound on $g_0 - w$; for $x < 0$ add (12). This is exactly (10). $\square$

4 Scope and novelty status

Theorem 1 is algebraic and retains all discontinuous quadratic solutions. Theorem 3 answers the canonical uniform Hyers–Ulam problem in Banach targets for every n and every nonzero parameter quadruple, including the resonant constant family when $c + d = 2$. Generalized Rassias controls and quasi- $(2, \beta)$ targets are separate refinements, not needed for the stated uniform result.

Exact-equation, absolute-radical, title, parameter, and Ulam-stability searches through 11 August 2026 found adjacent radical quadratic/quartic stability papers but no later source giving this all- n classification and uniform theorem. The packet should therefore be treated as a candidate new result until an expert completes a citation audit.

References

- [1] I. El-Fassi, *A new fixed point approach to hyperstability of radical-type functional equations in quasi- $(2, \beta)$ -Banach spaces*, arXiv:2007.01454v1, 2020.
- [2] J. Aczél and J. Dhombres, *Functional Equations in Several Variables*, Cambridge University Press, 1989.